MILITARY SPECIFICATION

CABLE AND CABLE ASSEMBLIES, SPECIAL PURPOSE,

ELECTRICAL, MAGNETIC MINESWEEPING (QUAD CABLE)

1. SCOPE

1.1 Scope. This specification covers quad type cable and quad cable assemblies for magnetic minesweeping.

1.2 Classification. Quad type magnetic minesweeping cables shall be classified as follows:

Type PQ1 - Four quadded, single conductor, insulated nonbuoyant cables.
Type PQ2 - Four quadded, single conductor, insulated nonbuoyant cables.
Type PQ3 - Four quadded, single conductor, insulated nonbuoyant cables.
Type AQ1 - Four quadded, single conductor, insulated buoyant cables.
Type AQ2 - Four quadded, single conductor, insulated buoyant cables.
Type AQ3 - Four quadded, single conductor, insulated buoyant cables.
Type S2 - Insulated, single conductor, buoyant.
Type S3 - Insulated, single conductor, buoyant.
Type K4 - Noninsulated, single conductor, nonbuoyant.

1.2.1 Standard lengths. Quad cables shall be furnished in the standard lengths shown in table I as specified (see 6.1).

Table I - Standard lengths.

Cable type	Length feet	Tolerance in percent of length	
		Plus	Minus
PQ1 quad section	1/ 70	5	0
PQ2 quad section	1/ 100	5	0
PQ3 quad section	1/, 2/ 130	5	0
AQ1 quad section	675	2	1
AQ2 quad section	675	2	1
AQ3 quad section	3/ 675	2	1
S2	900	2	2
S3	900	2	2
K4	150	2	2

1/ Two of the cables in a PQ quad section shall be sufficiently longer than the nominal length to permit staggering the PQ-AQ and AQ-AQ pigtail joints as specified by the referenced drawings.

2/ Two of the cables in the PQ3 quad section shall include, as a part of the specified length, 9 foot lengths of 2000 MCM flexible copper pigtails, and the other two shall include 10 foot lengths of 2000 MCM flexible copper pigtails.

3/ Exclusive of pigtails.

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1.2.2 Assemblies. Quad cable assemblies shall be of the following types as specified (see 6.1):

Q1 Quad assembly - One 70 foot PQ1 section, one 675 foot AQ1 section, and one 150 foot type K4 electrode.
Q2 Quad assembly - One 100 foot PQ2 section, one 675 foot AQ2 section, and one 150 foot type K4 electrode.
Q3 Quad assembly - One 130 foot PQ3 section, one 675 foot AQ3 section, and one 150 foot type K4 electrode.
Q2 Single assembly (S2 assembly) - One 900 foot length of type S2 cable and one 150 foot type K4 electrode.
Q3 Single assembly (S3 assembly) - One 900 foot length of type S3 cable and one 150 foot type K4 electrode.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
L-P-535 - Plastic Sheet (Sheeting); Plastic Strip; Vinyl Chloride Polymer and Vinyl Chloride-Vinyl Acetate Copolymer, Rigid.
PPP-B-1055 - Barrier Material, Waterproofed, Flexible.

MILITARY
MIL-G-160 - Grips, Cable (Naval Shipboard Use).

STANDARD

FEDERAL
FED-STD-601 - Rubber: Sampling and Testing.

DRAWINGS

MILITARY
S8101-1441851 - General Assembly - Type Q2 Cable.
S8101-1441852 - Assembly and Details of Type Q2 Cable, PQ2 Terminals.
S8101-1441853 - Type Q2 Cable - AQ2, S2 Terminal Assembly.
S8101-1441854 - Type Q2 Cable - AQ2, S2 Terminal Details.
S8101-1441856 - Q2 Design - Cross Section of Cables.
S8101-1449530 - General Assembly - Type Q3 Cable.
S8101-1449531 - Type Q3 Cable, Assembly of PQ3 Terminal.
S8101-1449532 - Type Q3 Cable, Details PQ3 Terminals.
S8101-1449533 - Type Q3 Cable, AQ3 - S3 Terminal Assembly.
S8101-1449534 - Type Q3 Cable, AQ3 - S3 Terminal Details.
S8101-1449535 - Q3 Cable, Cross Section of Cables.
S8101-1449537 - General Assembly - Type Q1 Cable.
S8101-1449536 - Assembly and Details of Type Q1 Cable - PQ1 Terminals.
S8101-1449539 - Type Q1 Cable, AQ1 - S2 Terminal Assembly.
S8101-1449540 - Type Q1 Cable, AQ1 - S Terminal Details.
S8101-1449541 - Q1 Cable, Cross Section of Cables.
S8101-1449611 - Assembly and Details - Type K4 Cable With Pressed End Fitting.
S8101-1535865 - Details of 9 Foot 6-Inch Diameter by 7 Foot T by 5-Foot 6-Inch Ø Shipping Reel for Q1-Q2 (Quad Sections) and S3 (Single Section).
S8101-1535866 - Details of 8-Foot Diameter by 4 Foot T by 4-Foot 8-Inch Ø Shipping Reel for S1-S2 (Single Section).
S8101-1535867 - Assembly and Details of 11-Foot Diameter, 8-Foot 4-Inch T by 6 Foot Ø Shipping Reel for Q3 (Quad Sections).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
E-42 - Operating Light and Water-Exposure Apparatus (Carbon Arc Type) For Artificial Weathering Test.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. Materials of construction shall be of the best quality and entirely suitable for the intended application.

3.1.1 Synthetic rubbers.

3.1.1.1 Insulation. Synthetic rubber compounds used in the insulation shall be butadiene-styrene copolymer (SBR), chlorosulphonated polyethylene, or a combination thereof having a resistivity of not less than 3×10^{14} ohm-cm. Insulation compounds shall vulcanize readily to compounds used in the cable jacket and to natural rubber.

3.1.1.2 Jacket. Synthetic rubber compounds used in the jacket shall be polychloroprene or chlorosulphonated polyethylene. They shall be tough and resilient and shall have high flexibility, tensile strength and resistance to ozone attack, sunlight, heat, water, oil and abrasion. They shall vulcanize readily to the synthetic rubbers used in the insulation and to natural rubber.

3.1.1.3 Insulation - jacket material combinations. Where insulation material is SBR, jacket material may be polychloroprene or chlorosulphonated polyethylene. Where insulation material is chlorosulphonated polyethylene or a combination of chlorosulphonated polyethylene and SBR, jacket material shall be chlorosulphonated polyethylene.

3.1.2 Reinforcing fabric or braid. Fabric or braid used to reinforce the synthetic rubber in the cable insulation shall be strong, flexible material made of natural or synthetic fibers. It shall be bonded firmly to the synthetic rubber materials and shall not be damaged by the temperatures used to vulcanized the rubber. Reinforcing fabric or braid shall not be used in the cable jacket.

3.2 General requirements. Cable assemblies shall be as specified hereinafter. In the event of any conflict between the requirements of this specification and the referenced drawings, the requirements of this specification shall govern.

3.2.1 No factory splices shall be allowed in individual sections of any cable, unless approved by the command or agency concerned.

3.2.2 Cable assemblies shall have high flexural endurance and shall be sufficiently flexible at all temperatures between -37° and 60°C to permit reeling on drums having diameters as follows:

Q1	quad assembly	42 inch diameter drum
Q2	quad assembly	66 inch diameter drum
Q3	quad assembly	72 inch diameter drum
S2	single assembly	66 inch diameter drum
S3	single assembly	72 inch diameter drum

3.2.3 Cable sheath. The cable sheath shall consist of an inner insulation in accordance with 3.1.1.1 and an outer jacket in accordance with 3.1.1.2. The insulation and jacket shall be vulcanized together in such a manner as to produce a single homogeneous sheath about the conductor and shall withstand the ply separation test specified in 4.4.5.3. The sheath of type PQ cables shall be suitable for transmitting the breaking stresses specified in 3.4.2.3, 3.5.3.3 and 3.6.3.3 to cable grips of the type specified in MIL-G-160.

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3.2.4 The sheath of flexible copper terminal - cables used on components of quad cable assemblies, shall be the same as that used on the associated cable component. The molded sheath over terminal lug barrels shall be the same material as is used for the cable jacket.

3.2.4.1 The insulation of the terminal lugs shall be securely vulcanized to the sheath of the terminal cables and firmly bonded to the terminal barrels and flanges. When tested in accordance with 4.4.6, the adhesion shall be not less than 10 pounds per inch of width.

3.2.5 Quad cables (PQ and AQ sections) shall be assembled with the exposed edges of sheath tapes facing away from the terminal end of the PQ section. "S" section electrodes shall be attached to the "S" section terminal which faces the exposed edges of sheath tapes.

3.2.6 The end faces of PQ terminal plugs shall be even within one quarter inch when PQ cable is lying on a flat surface.

3.2.7 Curing tapes shall be completely removed from the cable and from the central core of quad cable.

3.3 Government furnished equipment. When specified (see 6.1), the Government will furnish the reels specified in 5.2.1.

3.4 Type Q1 cable assemblies. Type Q1 cable assemblies (quad assembly and single assembly) shall be in accordance with Drawings S8101-1449537 through S8101-144 , inclusive and Drawing S8101-1449611.

3.4.1 Type PQ1 cable. Type PQ1 cable shall consist of four insulated nonbuoyant single-conductors cabled together to form a quad and shall be suitable for conducting direct current (d.c.) from the terminal box located on the deck of the ship to the type AQ1 cable.

3.4.1.1 Buoyancy. Type PQ1 cable shall be nonbuoyant.

3.4.1.2 Construction. Type PQ1 cable shall be constructed as shown on Drawing S8101-1449541.

3.4.1.2.1 Sheath thickness. The insulation and jacketing materials used in the sheath of type PQ1 cable shall have nominal thicknesses of 120 and 50 mils, respectively.

3.4.1.2.2 Terminals. The two shorter conductors of the type PQ1 quad section cable shall be terminated at one end by part 516 as shown on Drawing S8101-1449538. The two longer conductors of the PQ1 quad section shall be terminated at one end by part 517 as shown on Drawing S8101-1449538. The other ends of the four conductors shall be continuous with the type AQ1 cable as shown on Drawing S8101-1449538.

3.4.2 Type AQ1 cable. Type AQ1 cable shall consist of four insulated, self-buoyant, single-conductors cabled together to form a quad and shall be suitable for conducting d.c. from type PQ1 to type K4 and type S2 cable.

3.4.2.1 Buoyancy. Type AQ1 cable shall have a measured open end specific gravity of not more than 0.95.

3.4.2.2 Construction. Type AQ1 cable shall be constructed as shown on Drawing S8101-1449541.

3.4.2.2.1 Sheath thickness. The insulation and jacketing materials used in the sheath of AQ1 cable shall have nominal thicknesses of 120 and 50 mils, respectively.

3.4.2.2.2 Terminals. One end of the type AQ1 cable shall be continuous with the type PQ1 cable as shown on Drawing S8101-1449538. The PQ1 - AQ1 junction shall be marked with a band of red rubber not less than 4 inches in width. The other end of the type AQ1 cable shall be terminated as follows: Each conductor shall be terminated with a flexible connection consisting of (a) parts 527, 522 and 524, as shown on Drawing S8101-1449539; (b) 6 feet, 6 inches of 400 thousand circular mil (MCM), 61 by 7/.0306 inch medium hard drawn (MHD) copper cable and (c) part 520 as shown on Drawing S8101-1449539.

3.4.2.2.3 The two AQ1 conductors joined to the two shorter PQ1 conductors shall be joined together to form the short leg and the two AQ1 conductors joined to the two longer PQ1 conductors shall be joined together to form the long leg. The short and long legs shall be staggered 1 foot, 3 inches as shown on Drawing S8101-1449537.

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3.4.2.3 Physical and electrical properties. Type Q1 cable assemblies (quad assembly and single assembly) shall have the following physical and electrical properties:

(a) Current carrying capacity - 2200 amperes continuous with the buoyant section in sea water at a temperature of 85°F.
(b) Maximum d.c. resistance at 25°C. - total resistance of cable assembly excluding the K4 electrodes shall not exceed 0.044 ohm. (PQ1/AQ3 section 0.028 ohm, S2 section, 0.016 ohm.)
(c) Minimum breaking strength with terminals - 10,000 pounds for quad assembly, 15000 pounds for single - assembly.
(d) D.c. contact resistance between parts 518 and 517 and parts 518 and 516 as shown on Drawing S8101-1449538, and parts 524 and 522 and parts 523 and 559A as shown on Drawing S8101-1449539 shall be not greater than 4 microhms.

3.5 Type Q2 cable assemblies. Type Q2 cable assemblies (quad assembly and single assembly) shall be in accordance with Drawings S8101-1441851, S8101-1441852, S8101-1441853, S8101-1441854, S8101-1441856 and S8101-1449611.

3.5.1 Type PQ2 cable. The type PQ2 cable shall consist of four-insulated nonbuoyant single-conductors cabled together to form a quad and shall be suitable for conducting d.c. from the terminal box located on the deck of the ship to the type AQ2 cable.

3.5.1.1 Buoyancy. Type PQ2 cable shall be nonbuoyant.

3.5.1.2 Construction. Type PQ2 cable shall be constructed as shown on Drawing S8101-1441856.

3.5.1.2.1 Sheath thickness. The insulation and jacketing materials used in the sheath of type PQ2 cable shall have nominal thicknesses of 120 and 50 mils, respectively.

3.5.1.2.2 Terminals. The two shorter conductors of the PQ2 section shall be terminated at one end by part 556 as shown on Drawing S8101-1441852. The two longer conductors of the PQ2 section shall be terminated at one end by part 555 as shown on Drawing S8101-1441852. The other ends of the four conductors shall be continuous with the type AQ2 cable as shown on Drawing S8101-1441852.

3.5.2 Type AQ2 cable. The type AQ2 cable shall consist of four insulated, self-buoyant, single-conductors cabled together to form a quad and shall be suitable for conducting d.c. from type PQ2 to type K4 and type S2 cables.

3.5.2.1 Buoyancy. Type AQ2 cable shall have a measured open end specific gravity of not more than 0.95.

3.5.2.2 Construction. Type AQ2 cable shall be constructed as shown on Drawing S8101-1441856.

3.5.2.2.1 Sheath thickness. The insulation and jacketing materials used in the sheath of AQ2 cable shall have nominal thicknesses of 120 and 50 mils, respectively.

3.5.2.2.2 Terminals. One end of the type AQ2 cable shall be continuous with the type PQ2 cable as shown on Drawing S8101-1441852. The PQ2-AQ2 junction shall be marked with a band of red rubber not less than 4 inches in width. The other end of the type AQ2 cable shall be terminated as follows: Each conductor shall be terminated with a flexible connection consisting of (a) parts 564, 560 and 561 as shown on Drawing S8101-1441853, (b) 6 feet, 6 inches of 500 MCM, 61 by 7/.0342 inch MHD copper cable and (c) part 558 as shown on Drawing S8101-1441853.

3.5.2.2.3 The two AQ2 conductors joined to the two shorter PQ2 conductors shall be joined together to form the short leg and the two AQ2 conductors joined to the two longer PQ2 conductors shall be joined together to form the long leg. The short and long legs shall be staggered 1 foot, 3 inches, as shown on Drawing S8101-1441851.

3.5.3 Type S2 cable. Type S2 cable shall be a single conductor, self-buoyant, insulated cable suitable for conducting d.c. from type AQ1 (long leg) or type AQ2 (long leg) to type K4 cable.

3.5.3.1 Buoyancy. Type S2 cable shall have a measured open end specific gravity of not more than 0.95.

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3.5.3.2 Construction. Type S2 cable shall be constructed as shown on Drawing S8101-1441856.

3.5.3.2.1 Sheath thickness. The insulation and jacketing materials used in the sheath of the type S2 cable shall have nominal thicknesses of 120 and 50 mils, respectively.

3.5.3.2.2 Terminals. Each end of the type S2 cable shall be terminated with a flexible connection consisting of (a) parts 563, 559A and 523 as shown on Drawing S8101-1441853, (b) 6 feet, 0 inch of 1000 MCM, 61 by 7/.0484 inch MHD copper cable and (c) part 562 as shown on Drawing S8101-1441853.

3.5.3.3 Physical and electrical properties. Type Q2 cable assemblies (quad assembly and single assembly) shall have the following physical and electrical properties:

(a) Current carrying capacity - 5000 amperes on a 5-1/2 second on, 9-1/2 second off cycle with the buoyant section in sea water at a temperature of 85°F.
(b) Maximum d.c. resistance at 25°C. - total resistance of cable assembly excluding the K4 electrodes shall not exceed 0.04 ohm (PQ2/AQ2 section: 0.024 ohm, S2 section: 0.016 ohm).
(c) Minimum breaking strength with terminals - 15,000 pounds.
(d) D.c. contact resistance between parts 557 and 555 and parts 557 and 556 as shown on Drawing S8101-1441852, and parts 561 and 560 and parts 523 and 559A as shown on Drawing S8101-1441853 shall be not greater than 4 [illegible] ohms.

3.6 Type Q3 cable assemblies. Type Q3 cable assemblies (quad assembly and single assembly shall be in accordance with Drawings S8101-1449530 through S8101-1449535, inclusive and Drawing S8101-1449611.

3.6.1 Type PQ3 cable. The type PQ3 cable shall consist of four insulated, nonbuoyant, single-conductors cabled together to form a quad and shall be suitable for conducting d.c. from the terminal box located on the deck of the ship to the type AQ3 cable.

3.6.1.1 Buoyancy. Type PQ3 cable shall be nonbuoyant.

3.6.1.2 Construction. Type PQ3 cable shall be constructed as shown on Drawing S8101-1449535.

3.6.1.2.1 Sheath thickness. The insulation and jacketing materials used in the sheath of type PQ3 cables shall have nominal thicknesses of 120 and 80 mils, respectively.

3.6.1.2.2 Terminals. Two conductors of the PQ3 section shall be terminated with a flexible connection consisting of (a) parts 531 and 530 as shown on Drawings S8101-1449531 and S8101-1449532 (b) 9 feet, 0 inch of 2000 MCM, 61 by 19/.0415 inch alloy or tin coated copper cable and (c) part 528 as shown on Drawings S8101-1449531 and S8101-1449532. The other two conductors of the PQ3 section shall be terminated with a flexible connection consisting of (a) parts 531 and 530 as shown on Drawings S8101-1449531 and S8101-1449532 (b) 10 feet, 0 inch of 2000 MCM, 61 by 19/.0415 inch alloy or tin coated cable and (c) part 529 as shown on Drawings S8101-1449531 and S8101-1449532. These ends of the PQ3 conductors shall be staggered 1 foot as shown on Drawing S8101-1449530. The other ends of the four conductors shall be continuous with the type AQ3 cable.

3.6.2 Type AQ3 cable. The type AQ3 cable shall consist of four insulated self-buoyant, single-conductors cabled together to form a quad and shall be suitable for conducting d.c. from type PQ3 to type K4 and type S3 cable.

3.6.2.1 Buoyancy. Type AQ3 cables shall have a measured open end specific gravity of not more than 0.95.

3.6.2.2 Construction. Type AQ3 cable shall be constructed as shown on Drawing S8101-1449535.

3.6.2.2.1 Sheath thickness. The insulation and jacketing materials used in the sheath of the type AQ3 cable shall have nominal thicknesses of 120 and 80 mils, respectively.

3.6.2.2.2 Terminals. One end of the type AQ3 cable shall be continuous with the type PQ3 as shown on Drawing S8101-1449531. The PQ3-AQ3 junction shall be marked with a band of red rubber not less than 4 inches in width. The other end of the type AQ3 cable shall be terminated as follows: Each conductor shall be terminated with a flexible connection consisting of (a) parts 533, 534 and 535, as shown on Drawing S8101-1449533, (b) 8 feet, 6 inches of 1000 MCM, 61 by 7/.0484 inch MHD copper cable and (c) part 536 as shown on Drawing S8101-1449533.

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3.6.2.2.3 The two AQ3 conductors joined to the two shorter PQ3 conductors shall be joined together to form the short leg and the two AQ3 conductors joined to the two longer PQ3 conductors shall be joined together to form the long leg. The short and long legs shall be staggered 1 foot, 3 inches as shown on Drawing S8101-1449530.

3.6.3 Type S3 cable. The type S3 cable shall be a single conductor, self-buoyant, insulated cable suitable for conducting d.c. from type AQ3 (long leg) to type K4 cable.

3.6.3.1 Buoyancy. Type S3 cables shall have a measured open end specific gravity of not more than 0.95.

3.6.3.2 Construction. Type S3 cables shall be constructed as shown on Drawing S8101-1449535.

3.6.3.2.1 Sheath thickness. The insulation and jacketing materials used in the sheath of type S3 cables shall have nominal thicknesses of 120 and 80 mils, respectively.

3.6.3.2.2 Terminals. Each end of the type S3 cable shall be terminated with a flexible connection consisting of (a) parts 540, 539 and 538 as shown on Drawing S8101-1449533, (b) 7 feet, 4 inches of 2000 MCM, 61 by 19/.0415 inch MHD copper cable, and (c) part 537 as shown on Drawing S8101-1449533.

3.6.3.3 Physical and electrical properties. Type Q3 cable assemblies (quad assembly and single assembly) shall have the following physical and electrical properties:

(a) Current carrying capacity. 7500 amperes on a 5-1/2 second on, 9-1/2 second off cycle with the buoyant section in sea water at a temperature of 85°F.
(b) Maximum d.c. resistance at 25°C. Total resistance of cable assembly excluding the K4 electrodes shall not exceed 0.0195 ohm. (PQ3/AQ3 section: 0.0125 ohm, S3 section: 0.007 ohm).
(c) Minimum breaking strength with terminals. 15,000 pounds.
(d) D.c. contact resistance between parts 531 and 530 as shown on Drawing S8101-1449531 and parts 535 and 534 and parts 538 and 539 as shown on Drawing S8101-1449533 shall be not greater than 4 microhms.

3.7 Type K4 cable. Type K4 cable shall be single conductor nonbuoyant cable suitable for establishing an effective current path into or from the sea, using floats to suspend the cable.

3.7.1 Construction. Type K4 cable shall consist of 61 strands of annealed copper wire, each of 0.1071 inch nominal diameter, concentrically stranded. The cable shall be covered with two layers of fabric tape, each layer applied half-lapped, and No. 9 seine twine braid. Tape and braid of type K4 cables shall be rotproofed in accordance with Drawing S8101-1449611.

3.7.1.1 Terminals. Each end of type K4 cable shall be terminated with part 202 as shown on Drawing S8101-1449611.

3.7.2 Physical and electrical properties. Type K4 cable shall have the following physical and electrical properties:

(a) Minimum breaking strength with terminals. 10,000 pounds.
(b) Maximum d.c. resistance at 25°C. 0.0160 ohm per 1000 feet.

3.8 Cable sheath thickness. The minimum and maximum value of cable sheath thickness with reference to the nominal value shall be as follows:

(a) Average thickness of sheath shall be not less than 90 percent of the nominal value.
(b) Average thickness of sheath shall be not more than 125 percent of the nominal value. Buoyancy requirement shall not be changed.
(c) Any individual thickness shall be not less than 75 percent of the nominal value.
(d) The inside diameters of vulcanizing molds for cable terminations shall be the same as the diameter of the flanges of the respective termination fittings. This requirement is to permit the proper installation of waterproofing boots and sleeves.

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3.9 Identification.

3.9.1 Cable. The information listed in 3.9.1.1 shall be clearly molded into the insulation over the terminal barrels on both ends of type Q and type S cables. In addition, the serial numbers of type PQ, AQ and S cables shall be stamped into the ends of the terminals using steel stamps with 3/16 inch characters.

3.9.1.1 Identification information. Identification information shall be as follows:

(a) Cable type designation (for example, PQ2, AQ2, and S2).
(b) Manufacturer.
(c) Year of manufacture.
(d) Serial number.
(e) Federal stock number.

3.9.1.2 Serial numbers. Serial numbers will be furnished by the command or agency concerned.

3.9.2 Reels.

3.9.2.1 Identification plates. Both flanges of each reel of cable shall have securely affixed thereto identification plates made by heat sealing between two approximately 0.020 inch thick clear, transparent, vinyl sheets, in accordance with L-P-535 per printed with the marking information required (see 5.4). Identification plates of other types providing equivalent legibility and durability may be used when specifically approved by the command or agency concerned.

3.9.2.2 Color series. Each reel of cable shall be identified with a keyed series of colors to indicate the year of manufacture. This identification shall consist of a stripe approximately 2 inches wide and colored as specified herein for the year of manufacture. The cycle of colors shall be repeated every fifth year.

Year of manufacture	Identifying color
1969	Blue
1970	White
1971	Red
1972	Green
1973	Orange
1974	Blue

3.9.2.2.1 The colored stripe shall be applied circumferentially over the lagging (see 5.2.4) and midway between the flanges. The stripe shall consist of one coat, commercial quality, outside paint of appropriate color. In addition to the stripe, both flanges of each reel shall be stencilled with 4-inch high figures to show the year of manufacture.

3.9.2.3 Caution notice. The caution "DO NOT USE FORK LIFT" shall be painted conspicuously on each flange of each reel.

3.10 Workmanship. The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. An inspection lot shall consist of not more than 25 cables of one type.

4.2.2 One cable shall be selected at random from the first 6 production cables of a lot (or from the complete lot if the quantity is 6 or less) and the test samples specified in 4.2.2.1 or 4.2.2.2 taken from it. The cable shall then be reterminated. Removal of the test sample shall not reduce the length of the cable to less than the tolerance specified herein.

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4.2.2.1 Type AQ cables. Five feet shall be cut off the buoyant end of type AQ cables selected in accordance with 4.2.2 along with the associated flexible terminal cables, connections and lugs. The pair of conductors forming the short leg shall be separated from the pair of conductors forming the long leg. One of these pairs shall have the cut ends terminated with the same type of termination as already on the sample, except that the duplex connector shall remain separated. The samples so prepared shall be subjected to the tensile test specified in 4.4.3.1. The remaining pair of conductors and the terminal lugs shall be used in performance of the tests specified in 4.4.2, 4.4.4, and 4.4.6.

4.2.2.2 Type S cables. Ten feet of buoyant conductor, along with the associated flexible terminal cables, connectors and lugs, shall be cut from one end of type S cables selected in accordance with 4.2.2. Three feet of buoyant conductor shall be cut from the end of the 10 foot sample and used in performance of the tests specified in 4.4.2 and 4.4.4. The remainder of the sample shall be terminated on the cut end in the same manner as the original cable and subjected to the tensile strength test specified in 4.4.3.2. After completion of the test, one terminal lug shall be removed and used for the insulation adhesion test specified in 4.4.6.

4.2.2.3 In the event of failure of any of the samples to pass the test specified in 4.4.2, 4.4.3, or 4.4.4, one additional cable shall be selected at random from the same lot and duplicate samples taken. All of the conductor samples shall be subjected to the test(s) failed. (In the case of the quad tensile test both the pair of conductors forming the long leg and the pair forming the short leg, shall be tested.) Failure of any one of the samples from the second cable selected shall be grounds for rejection of the lot.

4.2.3 One 2-inch wide circumferential sample of cable sheath shall be taken from each end (except PQ-AQ connection) of every quad cable conductor, and two 2-inch wide circumferential samples of cable sheath shall be taken from each end of every type S cable, and shall be tagged with the serial number of the cable from which removed. (These samples are from the sheath removed from terminal installation.) All of these samples shall be used for the sheath thickness test specified in 4.4.5.1. The samples obtained from the first 6 production cables of a lot shall be used for the tests specified in 4.4.5.2, 4.4.5.3, 4.4.5.4 and 4.4.5.5.

4.3 Examination. Production cable, shall be examined to determine conformance with this specification as to dimensions, materials, workmanship and general construction. Total resistance measurements shall be measured carefully, using accepted standard procedures. After winding on shipping reels, 25 feet of the type PQ cable shall be unreeled and stretched out flat and the PQ terminals checked for evenness. PQ terminals shall be even within 1/4 inch when so examined.

4.4 Tests. The tests specified in 4.4.1 through 4.4.6 shall be conducted.

4.4.1 Electrical.

4.4.1.1 Dielectric test. A dielectric test on each completed cable assembly, or each cable (except type K4) when not included in an assembly shall be performed by immersing the cable in water for 4 hours, after which time it shall be sujected to a potential of 2,500 volts r.m.s (60 cycles per second) for 5 consecutive minutes between the conductor and ground. If the test should be interrupted, the total duration shall be increased 20 percent of the specified duration for each interruption. Any cable which fails to meet this requirement may be repaired and again tested.

4.4.1.2 Insulation resistance. At completion of the dielectric test specified in 4.4.1.1 and while the cable is still in the water, the insulation resistance shall be measured using a 500 volt megohmmeter. Insulation resistance corrected to a temperature of 60°F. shall be not less than 100 megohms as measured between the installed terminals and ground.

(Note: For quad assemblies, two measurements are required; one for the paralleled long leg conductors, and one for the paralleled short leg conductors.)

4.4.1.3 Contact resistance. A d.c. contact resistance test shall be made on each pressed connection of each complete cable to determine conformance with the resistance values specified in 3.4.2.3, 3.5.3.3 and 3.6.3.3. (4 microhms).

4.4.2 Buoyancy.

4.4.2.1 Open end specific gravity. A 2 to 3-foot sample of each type of buoyant cable selected from the first lot under each contract shall be floated in a tank of fresh

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water at room temperature and lead weights applied along the length until the whole sample just sinks. The open end specific gravity is then calculated from:

$$\text{Specific gravity} = \frac{\text{Wt. of cable in air}}{(\text{Wt. of cable in air}) + (\text{Wt. of lead in air} \times \frac{10.4}{11.4})}$$

4.4.2.2 The measured open end specific gravity of the buoyant cables shall be not more than 0.95.

4.4.3 Tensile strength.

4.4.3.1 Type AQ cables. The samples selected in accordance with 4.2.2.1 shall have the newly terminated ends connected by means of an equalizing bridle or yoke and tested as follows: Type Q1 shall have a tensile load of 5,000 pounds and type Q2 and type Q3 shall have a tensile load of 7,500 pounds applied between bridle or yoke and end fitting for a 4 hour period. There shall be no permanent damage or deformation of the cable after test.

4.4.3.2 Type S cables. The samples of type S cables (see 4.2.2.2) shall have a tensile load of 15,000 pounds applied between end fittings for a 4 hour period. There shall be no permanent damage or deformation of the cable after test.

4.4.4 Compression test. A sample of each type of buoyant cable (see 4.2.2.1 and 4.2.2.2) shall be subjected to the following compressive force values for a period of 4 hours, applied between two flat plates 6 inches in length:

Type of cable	Compressive force, pounds
AQ1	2,000
AQ2	2,500
S2	3,000
AQ3	3,000
S3	3,000

The test shall result in no permanent deformation or damage to the cable.

4.4.5 Sheath tests.

4.4.5.1 Thickness. At least five separate measurements shall be made on each of the sheath samples taken in accordance with 4.2.3. The average of all the measurements taken on all samples from each cable assembly shall be considered the average thickness. Where the materials used in the insulation and jacket differ, the requirements of this paragraph shall apply to the insulation and jacket individually. Tolerances specified in 3.8 shall apply.

4.4.5.2 Tensile strength. The jacket material of 5 sheath samples from each lot (see 4.2.3) shall be tested for tensile strength using method 4111 of FED-STD-601. One specimen from each sample shall be tested. The average tensile strength of the specimens shall be considered the tensile strength of the sheath and shall be not less than 1500 pounds per square inch (psi).

4.4.5.3 Ply separation test. Five sheath samples from each lot (see 4.2.3) shall be tested for strength of bond between adjacent plies of taped construction sheath or between successive layers of extruded sheath and between the insulation and jacket using method 4221 of FED-STD-601 except that the force shall be applied in such a manner as to tend to separate the individual layers of insulation and jacket along their mating faces. One specimen from each sample shall be tested. Bond between individual plies, jacket and insulation, or between either of these and an intervening reinforcing tape, braid or serving shall be not less than 10 pounds per inch of width.

4.4.5.4 Elongation test. Five sheath samples from each lot (see 4.2.3) shall be tested for elongation of jacket material using method 4121 of FED-STD-601. The jacket samples shall be obtained by slicing the jacket material from the sheath samples. One specimen from each sample shall be tested. Ultimate elongation shall be not less than 250 percent.

4.4.5.5 Oxygen and sunlight resistance. Jacket material from a minimum of 5 sheath samples from each lot (see 4.2.3) shall undergo an accelerated aging procedure consisting of the exposures specified in methods 7111 and 7311 of FED-STD-601. Twenty-four hours after accelerated aging is complete the tensile strength and elongation shall be not less

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than 60 percent of the value obtained in 4.4.5.2 and 4.4.5.4. Surface checks and cracks shall not be visible to the naked eye. One specimen from each sample shall be tested. As an alternative to test method 7311 for sunlight resistance of 100 hour exposure in type D test equipment in accordance with ASTM E-42 may be used employing a cycle of approximately 100 minutes of arc light exposure followed by approximately 18 minutes of exposure to arc light and water spray.

4.4.6 Terminal lug insulation adhesion test. A test specimen shall be prepared from one of each type of terminal lugs selected in accordance with 4.2.2.1, 4.2.2.2 and the adhesion of the insulation to the lug barrels tested by method 8031 of FED-STD-601. The sheath shall be reduced to a thickness of approximately 3/16 inch over the undistorted area of the lug during preparation of the specimen. Adhesion shall be not less than 10 pounds per inch of width.

4.5. Inspection of preparation for delivery. The packaging, packing and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.2)

5.1 One level of packaging and packing, as specified in 5.2 and 5.3 shall apply for all shipping destinations and all storage periods for cable covered by this specification.

5.2 Packaging.

5.2.1 Type PQ/AQ assemblies and S cables, with type K4 cables attached, shall be wound on separate reusable steel reels manufactured in accordance with Drawings S8101-1535865, S8101-1535866 and S8101-1535867, as applicable. "PQ/AQ" assemblies, with one K4 cable attached, shall be wound with the PQ terminals on top and "S" cables shall be wound with an attached K4 cable on the bottom. Both ends of the cable shall be securely fastened to the reel with rot resistance rope.

5.2.2 Type K4 cables, when procured separately, shall be wound on sturdy wooden shipping reels. Two type K4 cables shall be wound on each reel.

5.2.3 Type PQ cables, when procured separately shall be wound on sturdy shipping reels of appropriate diameter.

5.2.4 Lagging. The cable wound on reels shall be covered with flexible waterproofed wrapping paper conforming to PPP-B-1055 and secured to prevent unwrapping during handling and shipment. Reels shall be completely enclosed with wooden lagging with boards meeting each other. Lagging shall be 2-inch commercial lumber and shall be strapped with at least three 1-inch wide steel straps, except that reels containing K4 cables need have only two straps. Straps shall be stapled at intervals in accordance with the supplier's commercial practice. A minimum clearance of one cable diameter shall be provided between the inner face of the lagging and the waterproof wrapping material. When wooden reels are used the lagging shall be cut even with the outside of the reel flanges and nailed with cement coated steel nails. Lagging of steel reels shall be in accordance with the applicable steel reel drawing.

5.3 Packing. Cable packaged as specified in 5.2 shall not require overpacking. The method of packing or loading of reels shall be accomplished in a manner which will insure acceptance by carrier and will afford protection against physical or mechanical damage during shipment from the supply source to the using activity. The method of loading shall conform to the American Association of Railroads Rules Governing the Loading of Commodities and the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.4 Marking. In addition to any special marking required by the contract or order, each reel shall be marked as follows:

(a) Cable type.
(b) Cable serial number.
(c) Contract number.
(d) Manufacturer's name and destination.
(e) Federal stock number.

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5.4.1 The information specified in 5.4 shall be provided on identification plates (see 3.9.2.1).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of cable or cable assembly required (see 1.2.1 and 1.2.2).
(c) Whether the Government will furnish the reels specified in 5.2.1 (see 3.3).
(d) Lot quantity if special (see 4.2.1).
(e) Marking requirements other than those required by 5.4.

6.2 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.3 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "$" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
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APPENDIX

CORE CYLINDER FOR MAGNETIC MINESWEEPING CABLES

10. SCOPE

10.1 This appendix covers hollow plywood cylinders for the buoyant AQ and S sections, and solid wood cylinders for the nonbuoyant PQ sections of Q1, Q2 and Q3 magnetic minesweeping cable.

20. APPLICABLE DOCUMENTS

20.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

MMM-A-181 - Adhesive, Room-Temperature and Intermediate-Temperature Setting Resin (Phenol, Resorcinol, and Melamine Base).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. REQUIREMENTS

30.1 Dimensions and weights. The dimensions and weights of the hollow plywood cylinders shall be as follows:

Cable type	Outside diameter Inches	Length Inches	Shell thickness Inches	Head thickness Inches	Max. ave. wt. 300-1000 cylinders Grams
AQ1	1.55 ±0.020	2 ±0.030	0.110 ±0.015	0.250 ±0.035	21
AQ2	1.75 ±0.020	2 ±0.030	0.110 ±0.015	0.375 ±0.035	29
AQ3	2.40 ±0.025	2 ±0.035	0.110 ±0.015	0.375 ±0.035	51
S1-S2	2.50 ±0.025	2 ±0.035	0.110 ±0.015	0.375 ±0.035	54
S3	3.25 ±0.040	2.5 ±0.045	0.110 ±0.015	0.375 ±0.035	90

30.1.1 The dimensions of the solid wood cylinders shall be as follows:

Cable type	Diameter Inches	Length Inches
PQ1	1.12±0.010	1.50±1/16
PQ2	1.30±0.010	1.50±1/16
PQ3	1.78±0.015	2.0±1/16

30.2 Construction. Construction requirements shall be as specified in 30.2.1 through 30.2.7.2.

30.2.1 General. All cylinders shall be first class material and workmanship throughout.

30.2.2 Plywood cylinder heads. Cylinder heads shall be 5-ply pressure bonded poplar, gum or birch, or a combination thereof.

30.2.3 Plywood cylinder shells. Cylinder shells shall be 3-ply pressure bonded gum or birch. The plies shall be helically wound with adjacent plies in opposite directions. Veneer stock shall be of aircraft grade.

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30.2.4 Adhesive. Bonding of shell plies and between shell and heads shall be with a type I all-resorcinol adhesive conforming to MMM-A-181 applied and cured in accordance with manufacturer's detailed instructions.

30.2.5 Wood for solid cylinders. Wood from which solid cylinders are made shall be kiln dried, close grain birch, maple, sweetgum, American elm, beech or tupelo. There shall be no brash wood, shakes, knots, dots or any form of decay. (Mineral streak, stains and colorations shall not be considered as defects unless associated with decay.)

30.2.6 Construction detail.

30.2.6.1 Hollow plywood cylinders. Outer surface of cylinder heads shall be flush with ends of shell. The entire outer surface of the cylinder shall be sanded so as to provide a smooth base for the finish. Outer surface of cylinder heads shall be perpendicular to the cylinder axis, within a maximum deviation of 2 degrees.

30.2.6.2 Solid cylinders. The turned cylinder shall be sound, smooth, and free from wild grain. End surface of cylinder shall be perpendicular to the cylinder axis, within a maximum deviation of 2 degrees. Sharp edges shall be removed by turning a small radius at the ends of the cylinder.

30.2.7 Finish.

30.2.7.1 Hollow plywood cylinders. The entire outer surface of each cylinder shall be coated with a finishing material as follows:

First coat - Tumbled or dipped clear phenolic varnish.
Second coat - Tumbled, clear phenolic varnish.
Third coat - Tumbled clear phenolic varnish.

Additional color coding may be applied at the option of the cable manufacturer.

30.2.7.2 Solid cylinders. The entire surface of each cylinder shall be coated with a finishing material as follows:

First coat - Saturated tumbled clear phenolic varnish.
Second coat - Tumbled clear phenolic varnish.
Third coat - Tumbled clear phenolic varnish.

Color coding may be applied at the option of the cable manufacturer.

40. QUALITY ASSURANCE PROVISIONS

40.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

40.2 Tests.

40.2.1 Diameter check. The outer diameter of each cylinder shall be checked by means of "GO" "NO GO" gages. These are to consist of two 1/4 inch thick metal plates bored to the diameter listed below with the edges of the holes rounded.

Cable type	Nominal cylinder size Inches	GO gage Inches	NO GO gage Inches
AQ1	1.55	1.570	1.530
AQ2	1.75	1.770	1.730
AQ3	2.40	2.425	2.375
S1-S2	2.50	2.525	2.475
S3	3.25	3.290	3.210
PQ1	1.12	1.220	1.020
PQ2	1.30	1.310	1.290
PQ3	1.78	1.795	1.765

Cylinders shall pass through the GO gage and be unable to pass through the NO GO gage to be accepted.

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40.2.2 Construction check. The shell thickness and head thickness (plywood cylinders), length and construction detail shall be checked on one cylinder taken at random from each lot of 1000 cylinders to determine that the requirements of 30.1 and 30.2.6 have been met. If any cylinder fails to meet any of the requirements, five additional cylinders shall be taken at random from the same lot, and check made on the requirement or requirements which the original cylinder failed to meet. If any of these additional cylinders fail to meet the requirement or requirements in question, the lot of 1000 cylinders shall be rejected.

40.2.3 Weight check (plywood cylinders). A number (between 300 and 1000) of cylinders shall be taken at random from each lot of 100,000 plywood cylinders and weighed to determine that the maximum average weight requirement of 30.1.1 has been met.

40.2.4 Physical data (plywood cylinders). Fifteen cylinders shall be taken at random from the quantity of cylinders weighed in accordance with 40.2.3. These cylinders shall be used for the environmental resistance tests of 40.2.5. Prior to performing the environmental resistance tests the following data shall be taken and recorded:

(a) Diameter - Maximum and minimum, using diameter tape.
(b) Length - Maximum and minimum.
(c) Weight.
(d) Perpendicularity (see 30.2.6.1).

40.2.5 Environmental resistance tests.

40.2.5.1 Tap water immersion test. Three of the fifteen cylinders shall be immersed in tap water at room temperature for 100 hours and weighed (W_w). The cylinders shall then be subjected to 100°C. dry heat until 24 hour weight loss does not exceed 0.1 gram per cylinder and reweighed (W_D). Moisture absorption shall be calculated by the following formula and shall not exceed 25 percent; $\%MA = \frac{100W_W - W_D}{W_D}$

40.2.5.2 Cure test. Twelve of the 15 cylinders shall be exposed to a temperature of 100°F. and 1 0 percent relative humidity for 50 hours. Immediately following this exposure, they shall be sealed in a moistureproof container such as a rubber bag having a volume not more than 10 percent greater than that of the cylinders contained therein. They shall then be subjected to a temperature of 298°F. and a pressure of 70 p.s.i for a period of two hours. They shall then be air cooled for one hour at room temperature before opening the container. The difference in diameter at the center of the cylinder as measured by diameter tape before and after test shall not be more than 1.5 percent of the cylinder diameter before tests, and none of the cylinders shall collapse.

40.2.5.3 Radial crush test. Six of the cylinders precured as specified in 40.2.5.2 shall be subjected to a crush deformation test perpendicular to the axis of the cylinder. The deformation at loads tabulated below shall not exceed 5 percent of the nominal cylinder diameter.

Type	O.D. of cylinder Inches	Load in pounds
AQ1	1.55	800
AQ2	1.75	1400
AQ3	2.40	2000
S1-S2	2.50	2000
S3	3.25	2600

40.2.5.4 Boil delamination test. Three of the cylinders precured as specified in 40.2.5.2 shall be placed in boiling water for one hour and after removal shall show no evidence of delamination.

40.2.5.5 Failure to pass environmental resistance tests. If any cylinder fails to pass any of these tests, then a batch of 30 cylinders shall be selected at random from the same lot and subjected to the test which the original cylinders failed to pass. If more than 10 percent of the 30 cylinders fail to pass the test, the representative lot shall be rejected.

40.2.6 Test report. All test information shall be furnished to the prime contractor by the supplier.

50. PREPARATION FOR DELIVERY

50.1 Preparation for delivery shall be as specified by the cable manufacturer.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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